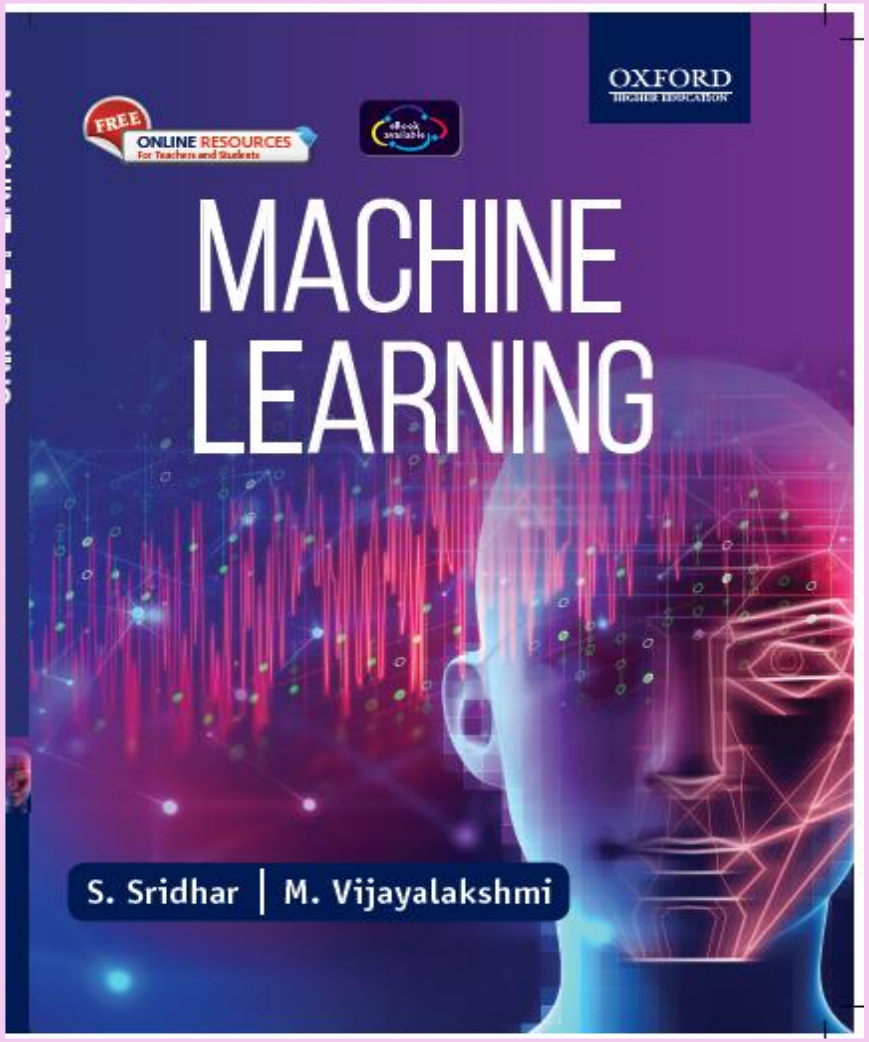


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Machine Learning

S. Sridhar and M. Vijayalakshmi

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Chapter 3

Basics of Learning Theory

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Learning

LEARNING IS A PROCESS BY WHICH ONE CAN ACQUIRE KNOWLEDGE AND CONSTRUCT NEW IDEAS OR CONCEPTS BASED ON THE EXPERIENCES.

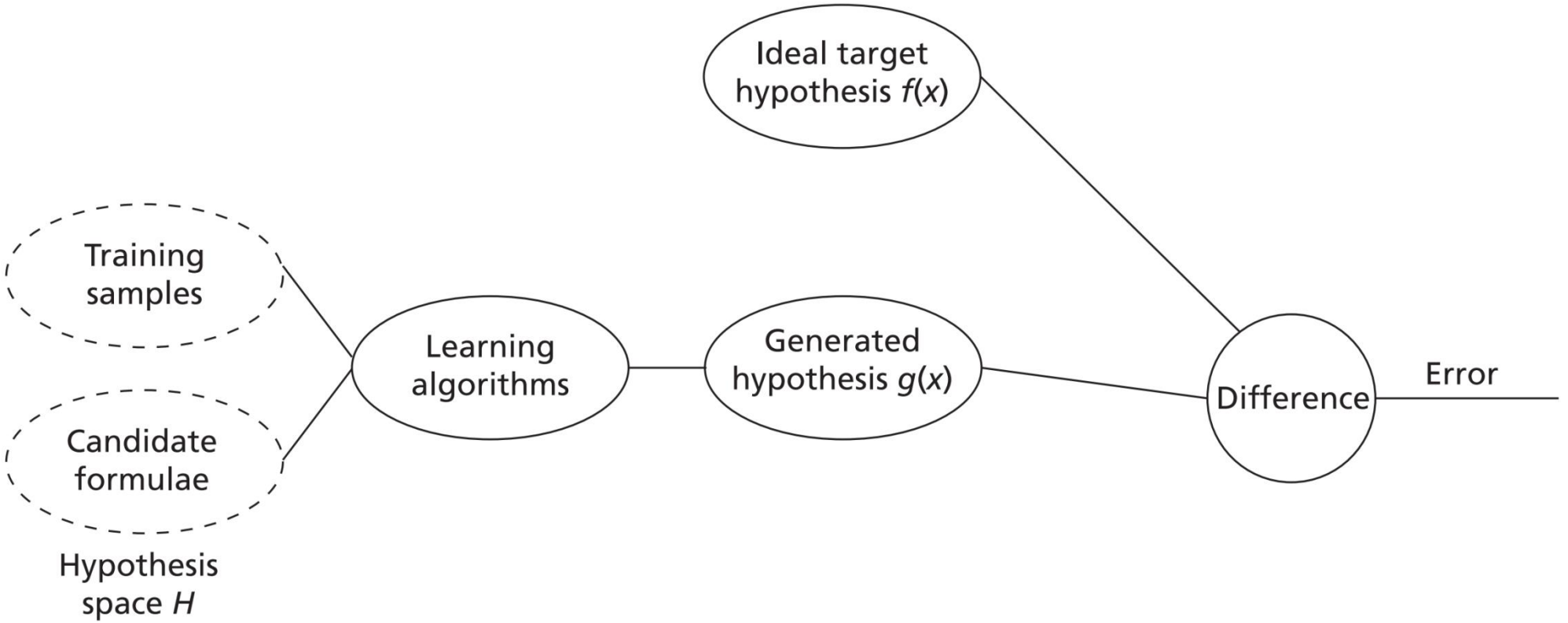
THE STANDARD DEFINITION OF LEARNING PROPOSED BY TOM MITCHELL IS THAT A PROGRAM CAN LEARN FROM *E* FOR THE TASK *T*, AND *P* IMPROVES WITH EXPERIENCE *E*.

THERE ARE TWO KINDS OF PROBLEMS – WELL-POSED AND ILL-POSED. COMPUTERS CAN SOLVE ONLY WELL-POSED PROBLEMS, AS THESE HAVE WELL-DEFINED SPECIFICATIONS AND HAVE THE FOLLOWING COMPONENTS INHERENT TO IT.

1. CLASS OF LEARNING TASKS (*T*)
2. A MEASURE OF PERFORMANCE (*P*)
3. A SOURCE OF EXPERIENCE (*E*)

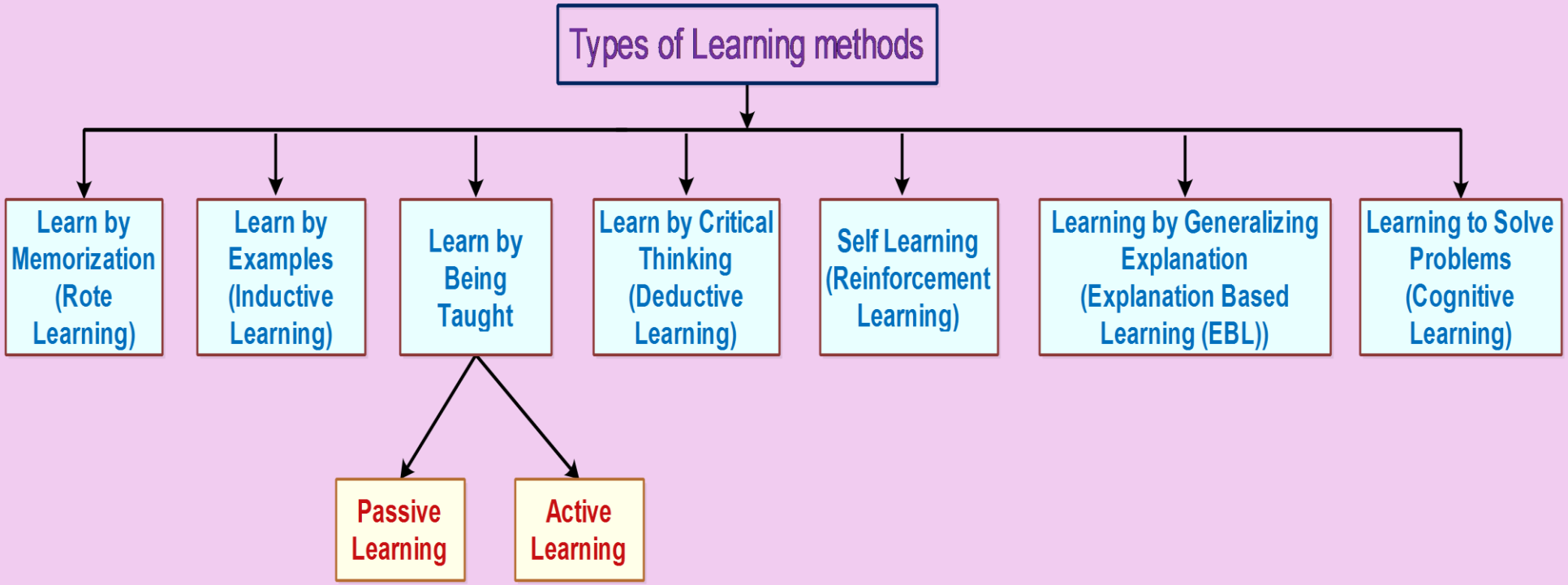
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Learning Environment



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Learning Types



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Introduction to Computation Learning Theory

There are many questions that have been raised by mathematicians and logicians over the time taken by computers to learn. Some of the questions are as follows:

1. How can a learning system predict an unseen instance?
2. How do the hypothesis *h* is close to *f*, when hypothesis *f* itself is unknown?
3. How many samples are required?
4. Can we measure the performance of a learning system?
5. Is the solution obtained local or global?

THESE QUESTIONS ARE THE BASIS OF A FIELD CALLED 'COMPUTATIONAL LEARNING THEORY' OR IN SHORT (COLT).

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Design of a Learning System

A system that is built around a learning algorithm is called a learning system. The design of systems focuses on these steps:

1. Choosing a training experience
2. Choosing a target function
3. Representation of a target function
4. Function approximation

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Introduction to Concept Learning

Concept learning is a learning strategy of acquiring abstract knowledge or inferring a general concept or deriving a category from the given training samples. It is a process of abstraction and generalization from the data.

Concept learning requires three things:

1. Input – Training dataset which is a set of training instances, each labeled with the name of a concept or category to which it belongs. Use this past experience to train and build the model.
2. Output – Target concept or Target function f. It is a mapping function $f(x)$ from input x to output y. It is to determine the specific features or common features to identify an object. In other words, it is to find the hypothesis to determine the target concept. For e.g., the specific set of features to identify an elephant from all animals.
3. Test – New instances to test the learned model.

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Representation of a Hypothesis

A *hypothesis* '*h*' approximates a target function '*f*' to represent the relationship between the independent attributes and the dependent attribute of the training instances. The hypothesis is the predicted approximate model that best maps the inputs to outputs. Each hypothesis is represented as a conjunction of attribute conditions in the antecedent part.

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Hypothesis Space

HYPOTHESIS SPACE IS THE SET OF ALL POSSIBLE HYPOTHESES THAT APPROXIMATES THE TARGET FUNCTION *F*.

THE SUBSET OF HYPOTHESIS SPACE THAT IS CONSISTENT WITH ALL-OBSERVED TRAINING INSTANCES IS CALLED AS **VERSION SPACE.**

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Heuristic Space Search

HEURISTIC SEARCH IS A SEARCH STRATEGY THAT FINDS AN OPTIMIZED HYPOTHESIS/SOLUTION TO A PROBLEM BY ITERATIVELY IMPROVING THE HYPOTHESIS/SOLUTION BASED ON A GIVEN HEURISTIC FUNCTION OR A COST MEASURE.

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Generalization and Specialization

SEARCHING THE HYPOTHESIS SPACE

THERE ARE TWO WAYS OF LEARNING THE HYPOTHESIS, CONSISTENT WITH ALL TRAINING INSTANCES FROM THE LARGE HYPOTHESIS SPACE.

1. SPECIALIZATION – GENERAL TO SPECIFIC LEARNING
2. GENERALIZATION – SPECIFIC TO GENERAL LEARNING

GENERALIZATION – SPECIFIC TO GENERAL LEARNING THIS LEARNING METHODOLOGY WILL SEARCH THROUGH THE HYPOTHESIS SPACE FOR AN APPROXIMATE HYPOTHESIS BY GENERALIZING THE MOST SPECIFIC HYPOTHESIS.

SPECIALIZATION – GENERAL TO SPECIFIC LEARNING THIS LEARNING METHODOLOGY WILL SEARCH THROUGH THE HYPOTHESIS SPACE FOR AN APPROXIMATE HYPOTHESIS BY SPECIALIZING THE MOST GENERAL HYPOTHESIS.

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Hypothesis Space Search by Find-S Algorithm

Algorithm 3.1: Find-S

Input: Positive instances in the Training dataset

Output: Hypothesis '*h*'

1. Initialize '*h*' to the most specific hypothesis.

   $h = <\varphi \quad \varphi \quad \varphi \quad \varphi \quad \varphi \quad \ldots..>$

2. Generalize the initial hypothesis for the first positive instance [Since '*h*' is more specific].
3. For each subsequent instances:

   If it is a positive instance,

   Check for each attribute value in the instance with the hypothesis '*h*'.

   If the attribute value is the same as the hypothesis value, then do nothing,

   Else if the attribute value is different than the hypothesis value, change it to '?' in '*h*'.

   Else if it is a negative instance,

   Ignore it.

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Limitations of Find-S Algorithm

1. Find-S algorithm tries to find a hypothesis that is consistent with positive instances, ignoring all negative instances. As long as the training dataset is consistent, the hypothesis found by this algorithm may be consistent.
2. The algorithm finds only one unique hypothesis, wherein there may be many other hypotheses that are consistent with the training dataset.

3. Many times, the training dataset may contain some errors; hence such inconsistent data instances can mislead this algorithm in determining the consistent hypothesis since it ignores negative instances.

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Version Spaces

The version space contains the subset of hypotheses from the hypothesis space that is consistent with all training instances in the training dataset.

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List-Then-Eliminate Algorithm

Algorithm 3.2: List-Then-Eliminate

Input: Version Space – a list of all hypotheses

Output: Set of consistent hypotheses

1. Initialize the version space with a list of hypotheses.
2. For each training instance,
   - remove from version space any hypothesis that is inconsistent.

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Candidate Elimination Algorithm

Algorithm 3.3: Candidate Elimination

Input: Set of instances in the Training dataset

Output: Hypothesis *G* and *S*

1. Initialize *G*, to the maximally general hypotheses.
2. Initialize *S*, to the maximally specific hypotheses.
   - Generalize the initial hypothesis for the first positive instance.
3. For each subsequent new training instance,
   - If the instance is **positive**,
     - Generalize *S* to include the positive instance,
       - Check the attribute value of the positive instance and *S*,
         - If the attribute value of positive instance and *S* are different, fill that field value with '?'.
         - If the attribute value of positive instance and *S* are same, then do no change.
     - Prune *G* to exclude all inconsistent hypotheses in *G* with the positive instance.
   - If the instance is **negative,**
     - Specialize *G* to exclude the negative instance,
       - Add to *G* all minimal specializations to exclude the negative example and be consistent with *S*.
         - If the attribute value of *S* and the negative instance are different, then fill that attribute value with *S* value.
         - If the attribute value of *S* and negative instance are same, no need to update '*G*' and fill that attribute value with '?'.
     - Remove from *S* all inconsistent hypotheses with the negative instance.

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Example 3.4:

Table 3.2: Training Dataset

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Logical Thinking	Interest	Job Offer
≥9	Yes	Excellent	Good	Fast	Yes	Yes
≥9	Yes	Good	Good	Fast	Yes	Yes
≥8	No	Good	Good	Fast	No	No
≥9	Yes	Good	Good	Slow	No	Yes

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Example 3.4: Deriving the Version Space

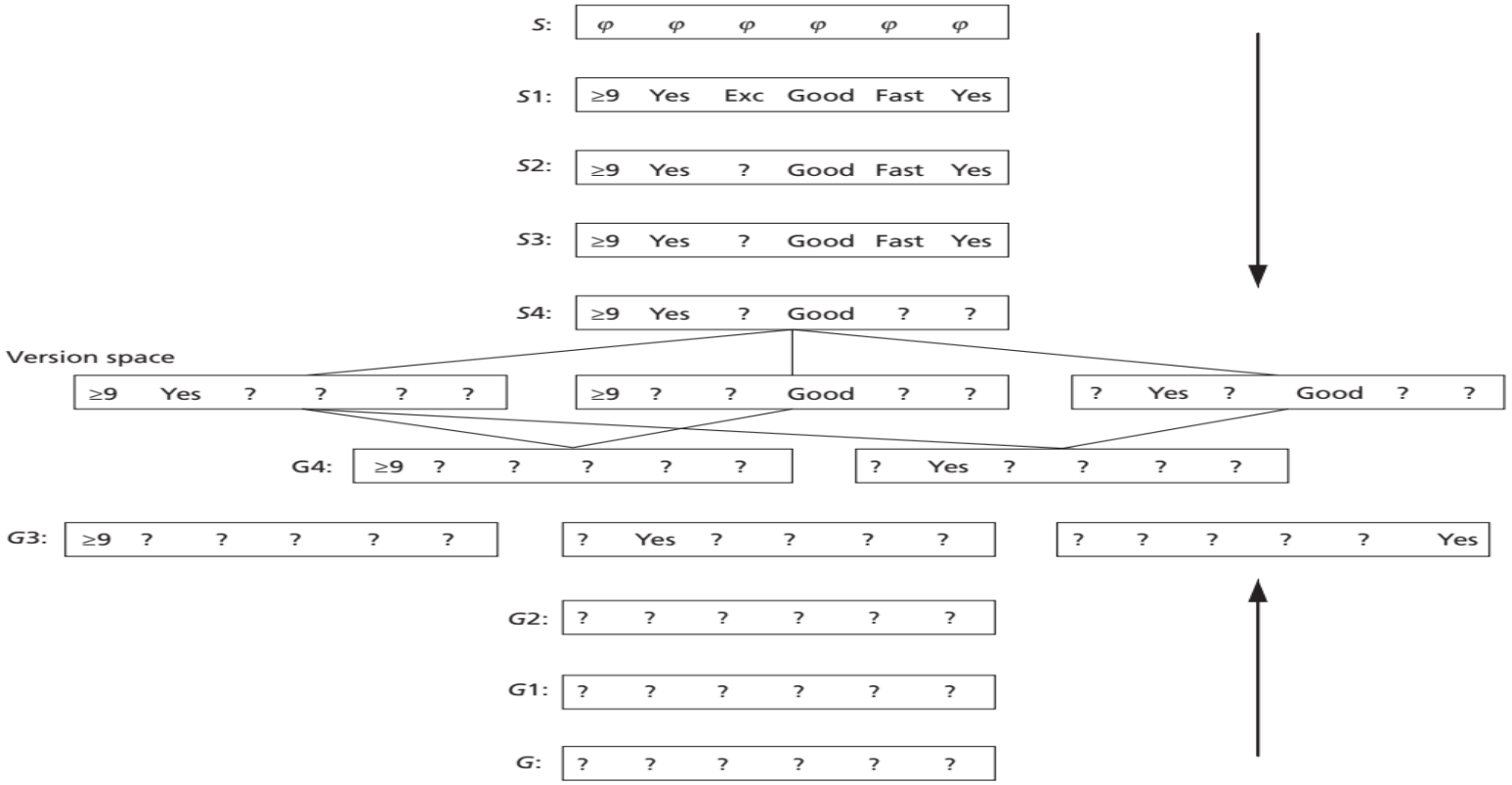


Figure 3.2: Deriving the Version Space

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Induction Biases

Induction is a process of learning a target function or generalizing training data into a general model. **Inductive bias** is the set of prior assumptions considered by a learning algorithm beyond the training data, in order to perform induction. It is also called as the ***bias*** of the algorithm. It is similar to prior knowledge used for learning new concepts.

There are two types of bias:

1. Constraint or Restriction – limit the hypothesis space
2. Preference – impose ordering on hypothesis space

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Bias and Variance

In supervised machine learning models, we need to find the target function $f(x)$ for an input value 'x' that best maps the predicted output values to actual output values. If the predicted output value deviates from the actual output value, then we call it an error. There are three kinds of prediction errors–Bias error, Variance error and Irreducible error.

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Bias vs. Variance

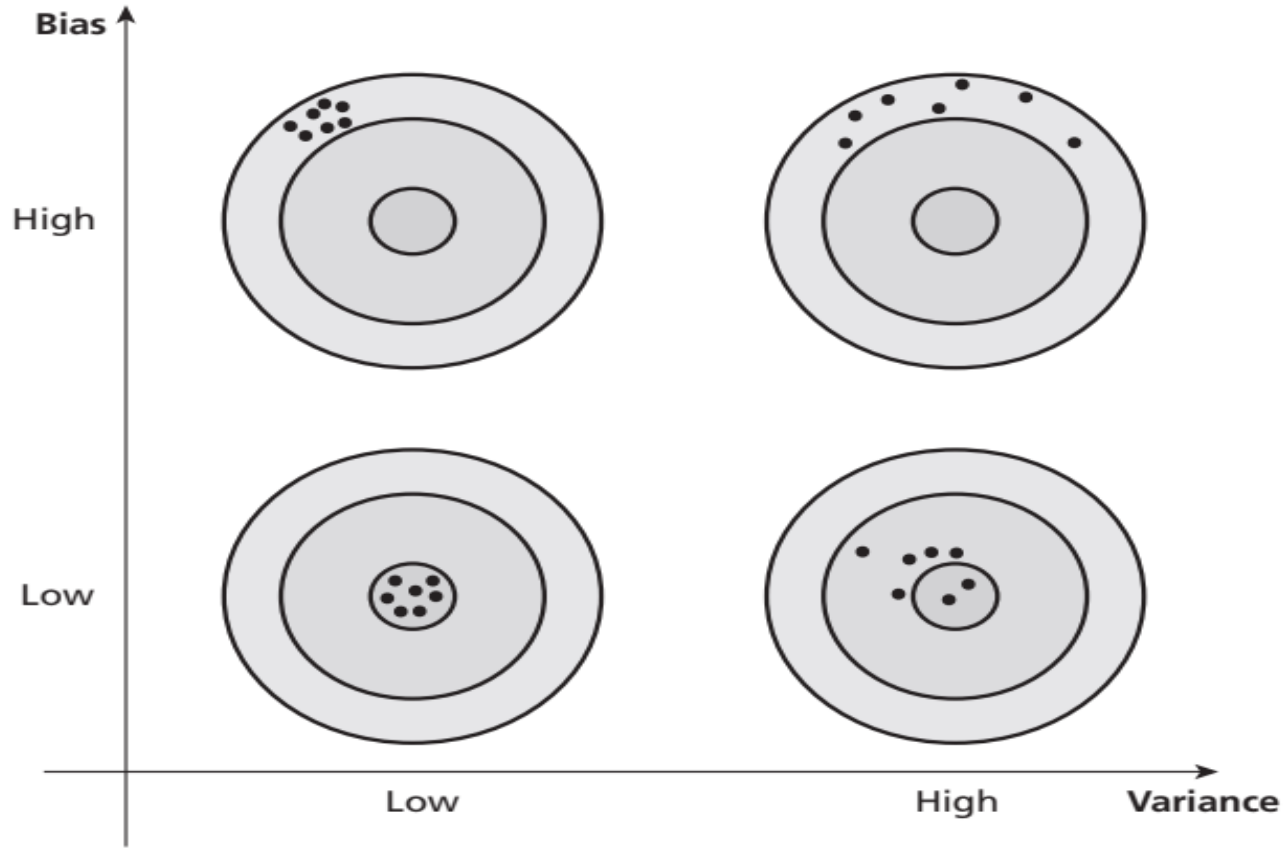


Figure 3.3: Bias vs Variance

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Bias vs. Variance Trade-off

Bias	Variance
Low Bias algorithms have less simplifying assumptions. Machine learning models: Decision Trees, k-Nearest Neighbors (k-NN) and Support Vector Machines (SVM)	**Low variance** occurs when training data is general. But low variance algorithms have a slight difference in the estimated target function when training dataset changes. Machine learning models: Linear Regression, Linear Discriminant Analysis (LDA) and Logistic Regression
High Bias algorithms have more simplifying assumptions or considers the features that are not relevant. They underfit with the training data. Machine learning models: Parametric algorithms, Linear algorithms – Linear Regression, Linear Discriminant Analysis and Logistic Regression, Naïve Bayes algorithm	**High variance** occurs when training data is more sensitive to outliers and noise. High variance algorithms have more difference in the estimated target function when the training dataset changes. Machine learning models: Non-linear algorithms - Decision Trees, k-Nearest Neighbors and Support Vector Machines, Non-parametric algorithms
Linear algorithms are fast learning due to simplifying assumptions and hence exhibit **high bias**.	Decision tree algorithms have **high variance**, when the tree is grown very deep and they are not pruned before use.
High bias algorithms are consistent but inaccurate on average.	**High variance** algorithms are accurate on average but are inconsistent.

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Modelling in Machine Learning

A machine learning model is an abstraction of the training dataset that can perform a prediction on new data. Training the model means feeding instances to the machine learning algorithm. Training datasets are used to fit and tune the model. After training a machine learning algorithm with the training data, a predictive model is generated as output to which a new data is fed to make predictions.

The process of modelling means training a machine learning algorithm with the training dataset, tuning it to increase performance, validating it and making predictions for a new unseen data. The major concern in machine learning is what model to select, how to train the model, time required to train, the dataset to be used, what performance to expect, and so on.

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Machine Learning Process

The four basic steps in the machine learning process are:

1. Choose a machine learning algorithm to suit the training data and the problem domain
2. Input the training dataset and train the machine learning algorithm to learn from the data and capture the patterns in the data
3. Tune the parameters of the model to improve the accuracy of learning of the algorithm
4. Evaluate the learned model once the model is built

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Re-sampling Methods

Re-sampling is a technique to select a model by reconstructing the training dataset and test dataset by randomly choosing instances by some method from the given dataset. This method involves selecting different instances repeatedly from a training dataset to tune a model. It is done to improve the accuracy of a model. The common re-sampling model selection methods are Random train/test splits, Cross-Validation (*K*-fold, LOOCV, etc.) and Bootstrap.

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Cross-Validation

Cross-Validation is a method by which we can tune the model with only training dataset. It is a model evaluation approach by which we can set aside some data of the training dataset for validation and fit the rest of the data to train the model. The best model is found by estimating the average of errors on different test data. The popular cross-validation family of methods includes Holdout method, *K*-fold cross-validation, Stratified cross-validation and Leave-One-Out Cross-Validation (LOOCV).

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Holdout Method

This is the simplest method of cross-validation. The dataset is split into two subsets called training dataset and test dataset. The model is trained using the training dataset and then evaluated using the test dataset. This holdout method can be applied for a single time which is called as single holdout method or it can be repeated for more than once which is called as repeated holdout method. The average performance on the test dataset is estimated to evaluate the model. Even though this model is very simple, it can exhibit high variance and the performance largely depends on how the dataset is split.

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K-fold Cross-Validation

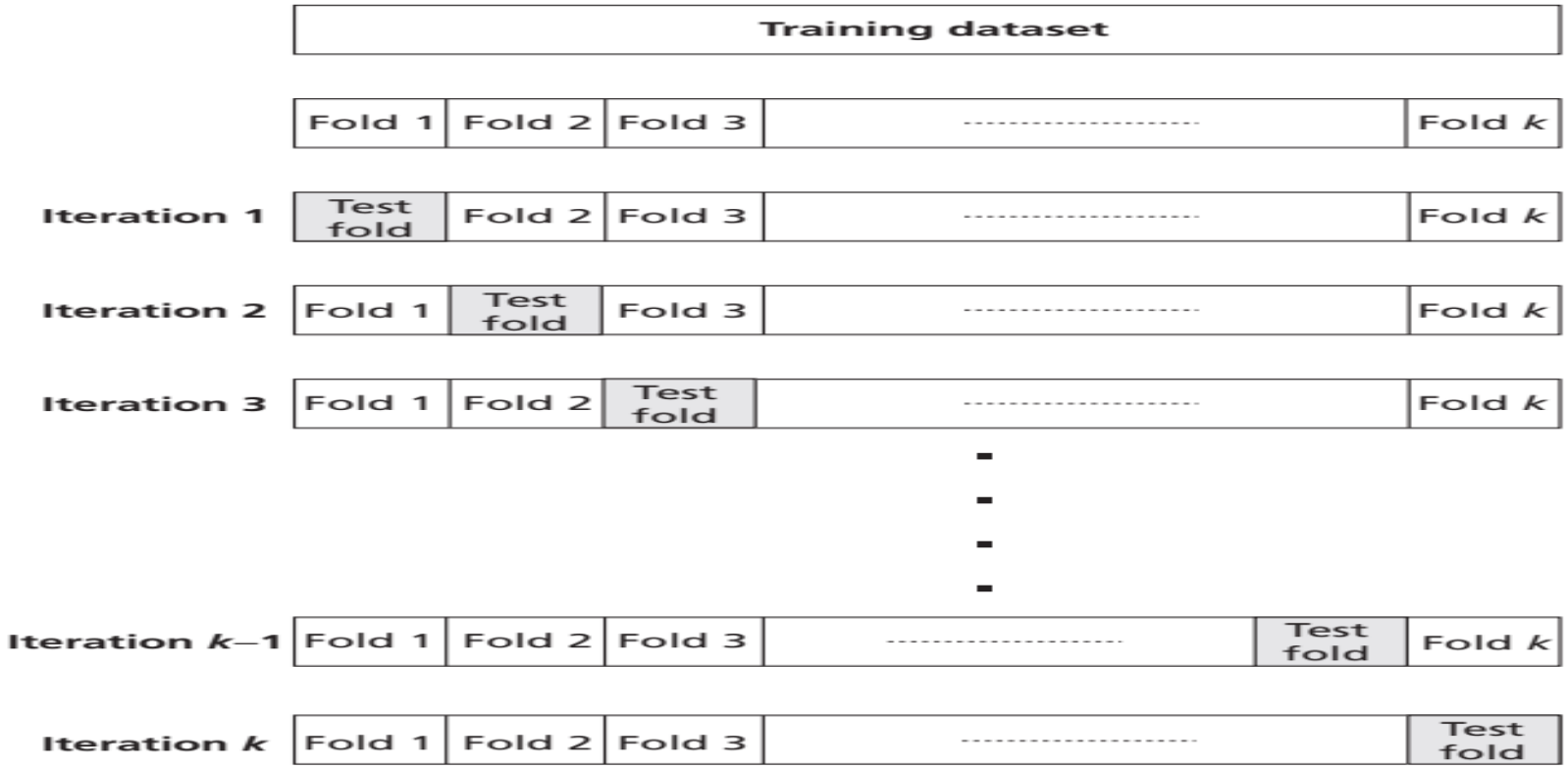


Figure 3.4: Illustration of *K*-fold Cross-Validation

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Stratified K-fold Cross-Validation

This method is similar to k-fold cross-validation but with a slight difference. Here, it is ensured that while splitting the dataset into k folds, each fold should contain the same proportion of instances with a given categorical value. This is called stratified cross-validation.

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Leave-One-Out Cross-Validation (LOOCV)

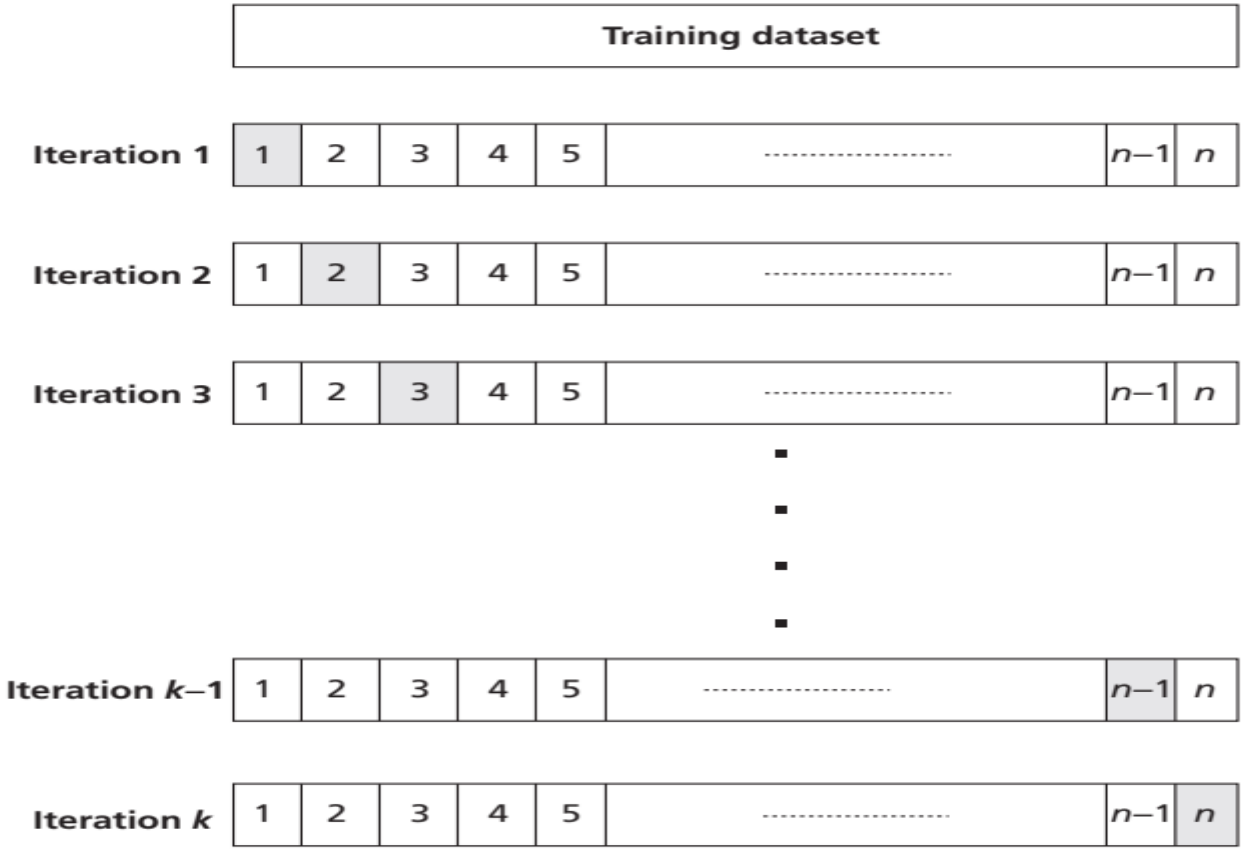


Figure 3.5: Illustration of Leave-One-Out Cross-Validation

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Model Performance

1. Sensitivity – The sensitivity of a test is the probability that it will produce a true positive result when used on a test dataset. It is also known as true positive rate. The sensitivity of a test can be determined by calculating:

$$\frac{TP}{TP + FN}$$

2. Specificity – The specificity of a test is the probability that a test will produce a true negative result when used on test dataset.

$$\frac{TN}{TN + FP}$$

3. Positive Predictive Value – The positive predictive value of a test is the probability that an object is classified correctly when a positive test result is observed.

$$\frac{TP}{TP + FP}$$

4. Negative Predictive Value – The negative predictive value of a test is the probability that an object is not classified properly when a negative test result is observed.

$$\frac{TN}{TN + FN}$$

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Model Performance

5. Accuracy – The accuracy of the classifier can be shown in terms of sensitivity computed as:

$$\frac{TP + TN}{TP + TN + FP + FN}$$

6. Precision – Precision is also known as positive predictive power. It is defined as the ratio of true positive divided by the sum of true positive and false positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision indicates how good classifier is in predicting the positive classes.

7. Recall – It is same as sensitivity.

$$\text{Recall} = \text{Sensitivity} = \frac{TP}{TP + FN}$$

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Visual Classifier Performance

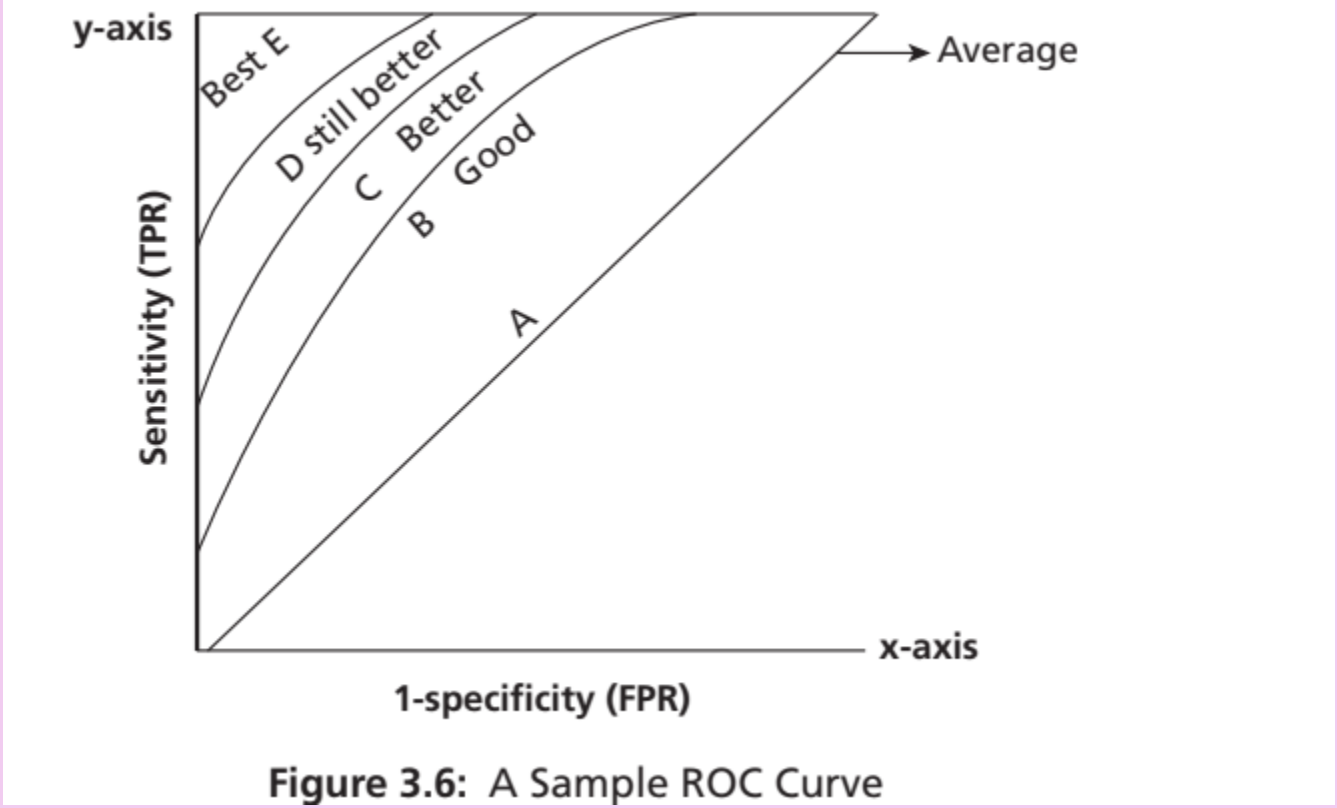


Figure 3.6: A Sample ROC Curve

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Scoring Methods

Another alternative for model selection is to combine the complexity of the model and performance of the model as a score. Then, model selection is done by selecting the model that maximizes or minimizes the score.

Minimum Description Length (MDL) is one such method. The aim is to describe target variable and model in terms of bits. MDL is the principle of using minimum number of bits to represent the data and model. It is a variant of Occom Razor's principle that states that the model with the simplest explanation is the best model. MDL too recommends the selection of the hypothesis that minimizes the sum of two descriptions of data and model.

Let h be a learning model. Let $L(h)$ is the number of bits used to represent the model and D is the number of predictions, then the MDL is given as:

$$L(h) + L(D \mid h)$$

where, $L(D \mid h)$ is the number of bits used to represent the predictions D based on the training set.

MDL can be expressed in terms of negative log-likelihood also as:

$$\text{MDL} = -\log(p(\Theta)) - \log(p(y \mid x, \Theta))$$

where, y is the target variable, x is the input and Θ is the model parameters.

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Learning Frameworks

PAC FRAMEWORK

PAC learning is a framework that performs a mathematical analysis of machine learning models in choosing the probably approximate correct hypothesis with low error. One of the important learning framework is called Probably Approximate Correct (PAC) framework. This framework was developed by Leslie Valiant as theoretical machine framework and deals with the quantification of difficulty in learning tasks.

The basic assumptions of PAC framework are given below:

1. The unseen samples are drawn from the past samples using a fixed distribution.
2. The rate of samples that are classified wrongly is called an error rate. Normally, Boolean functions with the loss rate on 0 – 1 can be considered as:

$$error(h) = Loss_{0/1}(h) = \sum_{x,y} L_{0/1}(y, h(x))p(x, y)$$

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Learning Frameworks

MISTAKE BOUND MODEL

Similarly, for Find S and Halving algorithm, mistake bound model can make at most $\lfloor \log_2 |H| \rfloor$ mistakes before exact learning. Then, for an arbitrary class C, the worst cast algorithm makes opt C mistakes, as:

$$\mathrm{VC}(C) \leq \mathrm{OPT}(C) \leq \log_2 (|C|)$$

where, VC(C) is the size of the largest subset that can be shattered by the hypothesis.

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Finite and Infinite Hypothesis

The *Sample Complexity* of a machine learning algorithm defines the number of training instances required to converge to a successful hypothesis for a target function. On the other hand, *Computational Complexity* defines how much computational effort is required for the machine learning model to converge to a successful hypothesis for a target function.

In a finite hypothesis space, PAC framework can identify the classes of hypotheses that can be learned from a polynomial number of training samples, whereas in an infinite hypothesis space, it cannot be learned from a polynomial number of training samples. Therefore, for an infinite hypothesis space, VC dimension defines the boundary for the number of training instances required for inductive learning.

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Estimating Hypothesis Accuracy

Evaluating the accuracy of the hypothesis '*h*' is very important in machine learning. There are some statistical methods to estimate the accuracy of the hypothesis. Sometimes, the hypothesis for a target function may outperform the training instances but when a test instance from the domain is given it would misclassify the instance. The *Sample Error* of the hypothesis is defined as the proportion of instances misclassified with the training instances. The *True Error* of the hypothesis is defined as the probability that the hypothesis will misclassify an instance drawn at random from the domain.

To estimate the True Error, confidence interval can be computed with the Sample Error, which would approximate it to 95%.

Given, n as the number of training instances and $error_s(h)$ as the Sample Error, confidence interval can be calculated as:

$$error_s(h) \pm 1.96\sqrt{\frac{error_s(h) - (1 - error_s(h))}{n}}$$

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Hoeffding's Inequality

Hoeffding's inequality gives an upper bound for 'h' in a finite hypothesis set as:

$$\mathrm{P}(|E_{\text{in}}(h) - E_{\text{out}}(h)| \geq \varepsilon\ ) \leq \delta = e^{-2n\varepsilon^2}$$

where, $\varepsilon > 0$ is some small value used to measure the deviation between E_{in} and E_{out}. However, this deviation is less than or equal to some bound δ which reduces exponentially as ε and/or the sample size n increases.

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Vapnik–Chervonenkis Dimension

With an infinite hypothesis set, a generalization bound can be derived using Vapnik–Chervonenkis (VC) dimension. In order to determine whether the learned function 'h' is PAC learnable and to derive the right bound, this VC dimension is required.

VC dimension of a classifier or a hypothesis 'h' is defined to be the maximum number of data points 'n' that can be shattered (classified/separated) into positive and negative data points for all possible combinations of the data points (i.e., to classify all possible 2^n labelling of data instances correctly).

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Summary

1. *Learning* is a process by which one can acquire knowledge and construct new ideas or concepts based on the experiences.
2. Concept learning is a learning strategy of acquiring abstract knowledge or inferring a general concept or deriving a category from the given training samples.
3. Each concept or category obtained by learning is a Boolean valued function which takes a true or false value.
4. A *hypothesis* '*h*' is an approximation of a target function '*f*' to represent the relationship between the independent attributes and the dependent attribute of the training instances.
5. The hypothesis is the predicted approximate model that best maps the inputs to outputs.
6. The most general hypothesis can allow any value for each of the attribute. It is represented as: <?, ?, ?, ?, ?, ?, ?, ?>.
7. The most specific hypothesis will not allow any value for any of the attribute. It is represented as, <φ, φ, φ, φ, φ, φ, φ, φ>.

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Summary

8. The hypothesis produced is also called as concept description which is a model that can be used to classify subsequent instances.
9. The set of all possible approximations of the target function is called as *Hypothesis space*.
10. The subset of hypothesis space that is consistent with all observed training instances is called as *Version Space*.
11. Hypothesis ordering is also important wherein the hypotheses are ordered from the most specific one to the most general, in order to restrict searching the hypothesis space exhaustively.
12. There are two ways of learning the hypothesis consistent with all training instances from the large hypothesis space called (a) general to specific, (b) specific to general.
13. Find-S algorithm is guaranteed to converge to the most specific hypothesis in H that is consistent with the positive instances in the training dataset.
14. List-Then-Eliminate Algorithm initializes the version space to contain all hypotheses and then eliminate any hypothesis that is found inconsistent with any training instance.
15. Candidate Elimination algorithm limits the version space to contain only those hypotheses that are most general and most specific.
16. Inductive Learning tries to induce a general function from specific training instances.
17. This way of learning a hypothesis that can produce an approximate target function with a sufficiently large set of training instances and can also approximately classify other unobserved instances is called as Inductive learning hypothesis.
18. There always exists a tradeoff between bias and variance.

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Summary

19. When a machine learning model learns too much from the training instances including noise, overfitting occurs.
20. Underfitting generally occurs when a machine learning model could not learn from the training instances or when predictors are very simple.
21. The process of modelling means training a machine learning algorithm with the training dataset, tuning it to increase performance, validating it and making predictions for a new unseen data.
22. Evaluating the selected machine learning model is also equally important as training the model.
23. Model selection is a process of selecting one good enough model among different machine learning models for the dataset or selecting different sets of features or hyperparameters for the same machine learning model.
24. Re-sampling is a technique to do model selection by reconstructing the training dataset and test dataset by randomly selecting instances by some method from the given dataset.
25. Cross-Validation is a method by which we can tune the model with only training dataset.
26. The popular cross-validation family of methods includes Holdout method, *K*-fold cross- validation, Stratified cross-validation and Leave-One-Out Cross-Validation (LOOCV).
27. When learning hypothesis, there is a need to measure the bias (error) between the target function f and the derived hypothesis h.
28. The relationship between the in-sample error and out-of-sample error can be quantified by using probabilistic bound expressed with a powerful tool called Hoeffding's inequality.
29. In an infinite hypothesis set, a generalization bound can be derived using *Vapnik–Chervonenkis (VC)* dimension.

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